

Ian Sebastian V. Macabulos

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PROJECTS

UA LabSign – University Laboratory Attendance | Next.js, TypeScript, PostgreSQL, React Native, Expo

- Engineered a web and Android attendance platform for the University of the Assumption, replacing manual paper-based laboratory attendance with centralized digital workflows designed to support hundreds of students, 20+ faculty members, and university administrators across multiple academic programs.
- Designed a secure, device-bound attendance protocol combining ECDSA P-256 signatures, GPS geofencing, institutional Google authentication, and expiring session PINs to verify user identity, authorized device, physical presence, and active class session before recording attendance.
- Architected synchronized Next.js and React Native/Expo clients backed by PostgreSQL and Prisma, supporting real-time attendance monitoring, class scheduling, report generation, device authorization and recovery, RBAC, and administrative audit trails.

AC-CORE – Municipal Maintenance and Incident Reporting System | Angular, Node.js, Express.js, MongoDB

- Engineered geospatial flood-risk routing across 14 high-risk neighborhoods using elevation data to identify potential downstream impact areas and support infrastructure prioritization.
- Developed a Paved Paradox prioritization model for road maintenance, combining infrastructure condition and commercial-area risk to surface higher-impact repairs during heavy rainfall.
- Built a responsive Angular Zoneless + Leaflet interface with marker clustering, GPS auto-detection, and a 30-second live work-order refresh for faster field reporting and monitoring.

TECHNICAL SKILLS

Frontend: Next.js, React, TypeScript, Tailwind CSS, Vue.js, Angular, React Native, Expo

Backend & Databases: Node.js, Nest.js, Express.js, FastAPI, PHP, Laravel, Java, Python, PostgreSQL, MongoDB

Tools & DevOps: Jest, Vitest, Playwright, Postman, Git, GitHub Actions, Vercel, AWS, Figma, WordPress

WORK EXPERIENCE

Backend AI Engineer Intern — FlyRank AI

- Engineered a TypeScript/Fastify usage metering and billing service for LLM applications, implementing tenant quotas, token tracking, idempotent requests, and micro-cent usage pricing with PostgreSQL.
- Integrated Stripe test-mode billing and webhook processing, with automated tests covering quota enforcement, usage accounting, checkout flows, and billing events.

Web Developer Intern — Sortr

- Redesigned and optimized WordPress production pages, improving responsive layouts, content structure, technical SEO, image delivery, and overall frontend performance.
- Built an internal IT service desk ticketing system that centralized hardware and software issue reporting, automated ticket routing and status tracking, and gave the IT team a structured workflow for managing support incidents from submission to resolution.

Freelance Full-Stack Developer

- Built and deployed custom full-stack systems for local businesses and institutional clients, replacing manual workflows with web applications for operations, reporting, attendance, and data management.
- Delivered projects across React/Next.js, Node.js, PostgreSQL, MongoDB, and cloud platforms, handling application architecture, database design, authentication, testing, deployment, and client handoff.

EDUCATION

Holy Angel University

2023 – Expected Graduation: 2027

Bachelor of Science in Information Technology, Major in Web Development

- Academic Honors:** Consistent Dean's and President's Lister, A.Y. 2023-2026
- Certifications & Training:** CompTIA IT Fundamentals (ITF+), AWS Cloud Foundations (AWS Academy), freeCodeCamp (Responsive Web Design, JavaScript Algorithms, Back End Development).